

Does Regional Economic Policy Uncertainty Matter to Cryptocurrency Returns? Evidence from China and Japan

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ABSTRACT

Increasing concerns about the impact of policy-related uncertainty on financial markets have drawn significant attention recently. This study provides new evidence on regional uncertainty spillovers in cryptocurrency markets by integrating time-varying Granger causality with machine learning algorithms. We show that Asian Economic Policy Uncertainty (EPU) correlation between China and Japan contains incremental predictive information for cryptocurrency returns beyond conventional global and the U.S. uncertainty measures. In contrast, EPU indices from other major Asian countries do not exhibit the significant predictive power. Out-of-sample results further indicate that the intermediate-term EPU correlation between China and Japan over a 180-day horizon delivers the strongest incremental forecasting improvement relative to baseline specifications. Our findings remain robust across alternative machine learning algorithms. The study contributes to expanding literature on digital asset markets by proposing a novel regional uncertainty co-movement measure to study regional EPU spillover effects and highlighting the importance of systematic policy regimes in shaping cryptocurrency return dynamics.

Keywords: Cryptocurrency; Spillover Effects; Economic Policy Uncertainty; Machine Learning; Granger Causality

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1. Introduction

As an emerging economic sector, the digital economy is technology-driven, distinguishing it from conventional economic activity (Ertz & Boily, 2019). In 2023, approximately half of the major contributors to the global digital economy were located in Asia (World Bank, 2023)¹. The region also leads the world in patent filings for innovative digital technologies, particularly artificial intelligence (World Bank, 2025). In policy uncertainty pricing models, uncertainty can raise required risk premia and generate predictable variation in asset returns (Pástor & Veronesi, 2013). Moreover, uncertainty shocks are often transmitted across economies, and regional spillovers may generate common uncertainty regimes that are more influential than country-specific disturbances (Biljanovska et al., 2021). We argue that cryptocurrency markets respond not only to idiosyncratic uncertainty shocks, but also to broader systemic policy regimes reflected in the correlation of the uncertainty across major economies.

China and Japan play a central role in Asia's macroeconomic and financial system, which motivates our focus on the co-movement of their EPU's. As the two largest economies in the region, accounting for roughly 56% of Asia's nominal GDP in 2024 (World Bank, 2024)², these two countries serve as major regional policy anchors, and policy developments in either country frequently generate spillovers across Asian financial markets (Biljanovska et al., 2021; IMF, 2024)³. Prior studies further show that uncertainty shocks originating from China and Japan contain distinct predictive information for cryptocurrency and broader asset markets (Cheng &

¹ The U.S., China, the U.K., Japan, Germany, and South Korea are the major contributors to the global digital economy. (World Bank, 2023).

² Based on the World Bank World Development Indicators (2024), China's nominal GDP in 2024 was approximately US\$18.3 trillion and Japan's was approximately US\$4.1 trillion. Together they accounted for roughly 56% of Asia's nominal GDP, making them by far the two largest economies in the region.

³ The IMF's Regional Economic Outlook: Asia and Pacific (October 2024) identifies China as the dominant source of macro-policy spillovers in the region. The Bank of Japan's monetary-policy normalization is highlighted as a separate, region-wide transmission channel.

Yen, 2020; Shaikh et al., 2020; Zhou et al., 2022). China and Japan are also the dominant players in the regional digital economy. The Boao Forum for Asia and CAICT (2023) estimate their combined digital-economy value about 77% of the US\$12.8 trillion aggregated across the fourteen leading Asian economies⁴. Given the growing importance of digital assets within Asia's increasingly integrated digital economy (World Bank, 2023), the interaction between China and Japan EPU may capture a broader regional uncertainty regime that is not reflected in country-specific uncertainty measures alone.

These considerations suggest that the co-movement of China and Japan policy uncertainty may contain information relevant for cryptocurrency pricing beyond conventional global uncertainty proxies. This argument is also consistent with the broader asset-pricing literature showing that policy uncertainty constitutes a priced source of systematic risk and generates predictable variation in asset returns (Pástor & Veronesi, 2013). Building on this premise, this study proposes a question, which we seek to address, does the time-varying co-movement of China and Japan EPU contain incremental predictive power for cryptocurrency returns?

Policy-related uncertainty may influence cryptocurrency markets through multiple channels. First, policy uncertainty pricing frameworks suggest that uncertainty constitutes a priced source of systematic risk, raising required risk premia and generating predictable variation in asset returns (Pástor & Veronesi, 2013). Second, policy uncertainty shocks affecting macro-financial conditions may undermine market participants' confidence in fiat-based monetary regimes, thereby increasing marginal demand for non-sovereign digital assets such as cryptocurrencies

⁴ Asian countries display imbalanced economic structures and uneven growth patterns. China dominates the global information and communication technology (ICT) market, with a 66.22% share in both manufacturing and service, 35.18% of the digital economy's labor force, and 26.28% of digital economy's exports. Both China and Japan demonstrate stronger productivity in the global innovation. China holds 58.53% of global ICT patent publications and Japan holds 8.4% in 2021. (World Bank, 2023).

(Eichengreen, 2019; Cheng & Yen, 2020). These theoretical insights offer a framework for examining the spillover effects of policy-related uncertainty on cryptocurrency returns.

Since the introduction of Bitcoin in 2009, cryptocurrency markets have expanded rapidly and become an important component of global digital finance (Bouri et al., 2017; Demir et al., 2018). This development aligns with broader trends emphasized by the World Bank regarding the growing role of digital financial innovation in the evolving digital economy (World Bank, 2023). As globally traded speculative assets, cryptocurrencies are highly sensitive to shifts in macro-financial conditions, investor sentiment, and time-varying risk premia (Demir et al., 2018; Negene & Wang, 2024), and they exhibit distinctive return dynamics and market structures (Wang, 2025; Wang et al., 2025).

In this study, we propose that regional policy-related uncertainty correlation between China and Japan may contain incremental predictive information for cryptocurrency returns beyond individual national EPU measures. However, the existing literature provides limited insight into the impact of regional uncertainty correlations on cryptocurrency markets, especially regarding the relative importance of co-movement between uncertainty factors. Some studies (Bouri et al., 2017; Anamika et al., 2023 and more) have demonstrated the spillover effects of EPU index, which is developed by Baker et al. (2016), has been widely used to measure the impact of economic policy uncertainty on various financial assets. Since cryptocurrency is a global digital asset (Wang et al., 2025), it is not only affected by policy-related uncertainty but also affected by global uncertainty factors such as the VIX (Malladi & Dheeriyaa, 2021), disease-related uncertainty (UI Haq, 2023), gold and oil volatility, and geopolitical uncertainty (Quan et al., 2020). Therefore, we attempt to fill this literature gap by examining whether regional EPU correlation of two largest Asian economies enhances the predictability of cryptocurrency returns. We further consider

uncertainty measures from other major Asian economies for comparison⁵. In addition, U.S. and global EPU indices are included as benchmark proxies to evaluate the incremental forecasting performance of our regional uncertainty measures.

To investigate the incremental predictability of uncertainty correlation between China and Japan, we classify the uncertainty factors into a benchmark (baseline) model and a regional (augmented) model. Our primary objective is to compare forecasting performance across model specifications. The baseline model includes global and U.S. uncertainty measures, whereas the augmented model consists of China and Japan EPU factors. To capture the systematic component of regional uncertainty synchronization, we construct rolling correlations in line with Forbes & Rigobon (2002) for the China and Japan EPU measures over short-, intermediate- and long-time horizons including 30-day, 60-day, 90-day, 120-day, 180-day, and 240-day periods. We evaluate whether augmenting the benchmark specification with regional correlation measures yields incremental benefits in return predictability. Model performance is assessed by comparing forecasting accuracy between the benchmark and augmented specifications.

To achieve our research objectives, we propose a two-stage empirical framework combining nonlinear Granger causality identification with machine learning prediction models (Wang, 2025) to assess the predictive importance of causality-based features. The first stage employs the nonlinear Granger causality Mackey–Glass (M–G) model to detect directional predictive linkages among variables, providing an economically interpretable foundation that standard machine learning approaches typically do not deliver. The second stage applies the Light Gradient Boosting Machine (LightGBM) algorithm to capture complex nonlinear forecasting

⁵ We include other major Asian countries such as India, South Korea, and Singapore in this study. However, we don't find evidence to show their causality relationships with cryptocurrencies.

relationships, while SHapley Additive exPlanations (SHAP) quantify the marginal contribution of each predictor and enhance model transparency (Giudici, 2018; Bussmann et al., 2021). By combining causality analysis with interpretable machine learning algorithms, it offers a broadly framework for studying dynamic relationships and return predictability across a wide range of financial markets. We then utilize dependence plots to illustrate how individual uncertainty factors influence cryptocurrency returns. Finally, we conduct robustness tests using alternative machine learning algorithms to verify the stability of our results.

Our out-of-sample results indicate that the regional EPU correlation between Japan and China over a 180-day horizon provides incremental predictive power for cryptocurrency returns beyond conventional uncertainty measures. Notably, this intermediate-term correlation appears to contribute most strongly to return predictability in cryptocurrency markets. However, we find no evidence of significant causal linkages between cryptocurrencies and uncertainty indices from other major Asian economies. The results further suggest that the VIX, the China–Japan EPU correlation (CHEJEP), oil prices (OIL), gold prices (GOLD), and geopolitical risk (GPRD) constitute the dominant uncertainty factors and key transmitters of return dynamics in cryptocurrency markets. Moreover, SHAP-based interpretations reveal that the predictive contributions of VIX, CHEJEP, OIL, CHE and GPRD are asymmetric across market conditions: their explanatory power strengthens during downturns but weakens when these indicators increase. By integrating time-varying Granger causality analysis with machine learning techniques, we provide economically meaningful insights for both market participants and policymakers.

This study contributes to the literature in several dimensions. First, to the best of our knowledge, this is the first study to examine the spillover effects of regional EPU correlation between China and Japan on cryptocurrency returns. By focusing on the time-varying co-

movement of uncertainty across two major Asian macro-policy anchors, our findings extend the emerging literature on cryptocurrency spillovers and demonstrate that correlated regional uncertainty provides incremental predictive power beyond conventional global uncertainty indicators. Second, we contribute methodologically by incorporating machine learning techniques to uncover nonlinear and previously hidden relationships between uncertainty correlations and cryptocurrency return dynamics. Third, we extend a two-stage empirical framework (Wang, 2025) that integrates time-varying Granger causality analysis with machine learning models. This strategy helps address interpretability concerns and allows us to evaluate linkages between uncertainty and cryptocurrency from both statistical and causal perspectives, thereby enhancing transparency in forecasting applications. Finally, our results have important implications for both academia and practice. Policymakers should recognize that regional uncertainty spillovers may transmit into cryptocurrency markets, highlighting the need for regulatory awareness and risk monitoring in digital financial systems. For researchers, our study motivates further interdisciplinary work and methodological advances to better understand the role of regional policy regimes in shaping cryptocurrency market behavior.

The remainder of the paper is organized as follows: Section 2 presents the literature review, Section 3 explains the data and methodology, Section 4 discusses the empirical results, and Section 5 concludes.

2. Literature Review

The notion that instable policy-related uncertainty may induce reallocation away from fiat-based assets toward alternative stores of value can be traced to the monetary substitution and currency-substitution literature (Friedman, 1956; Calvo & Vegh, 2010). More recently, the flight-

from-fiat mechanism has been extended to cryptocurrency, which are viewed as non-sovereign alternatives in settings of declining trust in traditional investment regimes or weakened policy credibility (Eichengreen, 2019). Consistent with this growing perception, the cryptocurrency market expanded rapidly, with total market capitalization roughly doubling in 2024 to approximately USD 3.26 trillion. As Bitcoin dominates the cryptocurrency market (Wang & Ngene, 2020), it accounted for more than 48.6% of total cryptocurrency market value in 2024⁶. In addition, regulatory developments have further underscored the growing role of cryptocurrencies in the global financial system. In January 2024, the U.S. Securities and Exchange Commission approved the first spot Bitcoin exchange-traded funds (ETFs), followed by the approval of spot Ethereum ETFs in July 2024. These developments have drawn increasing attention from scholars, practitioners and policymakers seeking to better understand how uncertainty shocks transmit into cryptocurrency markets.

However, the majority of academic studies still focus solely on Bitcoin, investigating the market efficiency (Urquhart, 2016; Bariviera, 2017; Nadarajah & Chu, 2017; Tiwari et al. 2018; Jiang et al., 2018), price bubbles (Cheah & Fry, 2015; Corbet et al., 2018), determinants of Bitcoin prices (Kristoufek, 2013; Glaser et al., 2014; Garcia et al., 2014; Hayes, 2017; Ciaian et al., 2016), and return volatilities (Bouoiyour et al., 2016; Dyhrberg, 2016; Paraskevi, 2017). Although Bitcoin is still the largest cryptocurrency by market capitalization⁷, other cryptocurrencies, such as ETH and stablecoin, have significantly gained traction, now comprising 50% of the total market, and actively trade in the market. Thus, it is crucial to broaden the scope of research to a broader cryptocurrency market to better understand the driving factors of the entire market (Sovbetov, 2018).

⁶ CoinMarketCap, (2024).

⁷ <https://coinmarketcap.com/>

Recent studies document the heterogeneous impact of EPU on cryptocurrencies across different countries (Haq et al., 2021; He et al., 2024). China EPU captures distinct predictive content for Bitcoin returns and volatility that is not replicated by U.S. or aggregate uncertainty measures (Cheng & Yen, 2020) and Bitcoin returns respond more actively to EPU in US, China and Japan (Shaikh et.al, 2020). Moreover, a growing body of work documents that EPU in major Asian economies, especially China and Japan, generates cross-border spillovers to global financial markets (Zhou et al., 2022; Li et al., 2023; Hoque et al., 2023; Osei et al., 2021), which could affect investor demand for cryptocurrencies as diversification or hedging tools. When these EPUs move together, we infer that regional policy conditions produce a systematic shift, adding incremental predictive power in cryptocurrency return beyond US EPU and global uncertainty. Our study fills the blank area in the digital economy in terms of evaluating the predictive power of the regional EPU co-movement between China and Japan. Our focus is motivated by the spillover effects transmitted to cryptocurrencies from across-country EPUs in a region.

Recent post-2023 studies further confirm that uncertainty effects are not uniform across countries and digital assets. Chowdhury and Abdullah (2024) document that cryptocurrency returns respond significantly to EPU, VIX, and macroeconomic variables, and Almeida and Gonçalves (2024) show that country-level EPU relationships vary across COVID-related regimes and investment horizons. Zhang et al. (2024) further demonstrate that global EPU affects cryptocurrency return stability, but this relationship weakens when cryptocurrency liquidity volatility is intensive. More recently, Civelli and Jackson (2025) document substantial heterogeneity across cryptocurrencies and show that cryptocurrency assets can hedge U.S. EPU while displaying safe-haven characteristics against Chinese EPU. These studies reinforce the need to examine market-wide cryptocurrency returns and regional EPU co-movement.

While extensive research examines the spillover effects of global risk factors, relatively little attention has been paid to how regional uncertainty influences cryptocurrency markets. Bitcoin reacts to global uncertainty, proxied by the first principal component of the VIXs of 14 developed and developing equity markets (Bouri et al., 2017). A higher value of VIX prompts investors to invest in digital currencies (Anamika et al., 2023). The global geopolitical risks (GPR) can predict the Bitcoin returns (Aysan et al., 2019). When geopolitical uncertainty reaches a higher level, the public tends to be less optimistic about the outlook of the economy and purchase more Bitcoin to avoid the GPR, thus driving the price of Bitcoin up (Su et al., 2020). A strong positive connectedness between GPR and Bitcoin returns is documented during the short term (Singh et al., 2022). Gold, a traditional safe-haven asset, also shows correlates positively with the cryptocurrency policy and price uncertainty indices (Hassan et al., 2021). Researchers find that oil price shock has a preponderant influence in determining cryptocurrency market volatility (Yin et al., 2021). Variations in oil prices could affect cryptocurrency market volatility through macroeconomic channels because they significantly co-vary with important macroeconomic indicators, such as inflation, real output, and interest rates.

Unlike other uncertainty indices that focus only on a group of developed economies, the World Uncertainty Index (WUI) measures uncertainty across a broad set of both developed and developing countries (Ahir et al., 2022). Liu & Gao (2002) observed the predictability power of WUI on the growth rate of the U.S. gross domestic product, indicating WUI as an important predictor for the real economy and market dynamics. The asymmetric impacts of WUI on the cryptocurrency portfolio's returns and liquidity are uncovered (Quang et al., 2019). Furthermore, infectious disease-related market volatility index (IDV) is proved by the existing studies that it can

impact the cryptocurrency markets (Haq et al., 2023). IDV is a common measurement for the extent of Covid-19 and other infectious diseases in influencing U.S. stock market volatility.

The findings about the heterogeneous actions of cryptocurrencies on EPU across different countries are mixed. Cheng & Yen (2020) showed that China EPU can predict bitcoin returns. They also demonstrate that U.S. EPU and other Asian EPU demonstrate weak predictability. However, Shaikh et.al (2020) document that EPU in the United States, China, and Japan possesses significant predictive power for Bitcoin returns. Regional EPU intensifies concerns about both regional and global economic growth⁸, exemplified by regional events such as a housing market collapse in China, monetary policy adjustments in Japan. A country's EPU cannot be completely isolated from global economic influences. In particular, countries within the same region tend to exhibit closer economic and policy linkages than other countries. But the regional EPU correlation effects between China and Japan has not been identified in the current cryptocurrency studies (Panagiotidis et al., 2019). The empirical evidence on the predictability of U.S. EPU for cryptocurrency markets remains controversial. Demir et al. (2018) found that U.S. EPU can negatively predict daily Bitcoin returns, but the impact turns out to be positive at the lower and higher quintiles of EPU. Wu et al. (2019) found Bitcoin reacts more to U.S. EPU shocks, while Wang et al. (2019) observed the risk spillover effect from U.S. EPU to Bitcoin is negligible.

Following the existing literature, we adopt a set of machine learning algorithms in this study. LightGBM is used to forecast cryptocurrency price trend (Sun et al., 2020), crude oil price

⁸ The expectation of global economic growth in the medium-term remains well below the pracademic historical average (IMF, 2024)⁸, intensifying the concerns about a declining global economy's growth. A slowdown in global economic growth could amplify the extent of EPU and worldwide uncertainty. Without promising economic prospects, firms can reduce or delay their investments, leading to a contraction in overall economic activities (Eberly, 1994). Uncertainty can significantly impact the economy through various channels i). increasing human capital costs (Schaal, 2017; Valletta & Bengali, 2013); ii). discouraging cross-border reallocation of capital and labor (Bloom et a. 2018); iii). reducing outputs in most large developed and developing economies at a country level (Kang et al. 2020; Fang et al. 2022); iv). adversely affecting spending decisions, as uncertainty diminishes stakeholders' confidence levels and prompts delayed purchases (Eberly, 1994).

(Jabeur et al., 2021), competition (Chiew and Choong, 2022), oil futures prices (Gong et al., 2022), and corporate financing risk (Wang et al. 2022), LightGBM based on the Gradient Boosted Decision Trees allows a large set of uncertainty factors. LightGBM tackles the tradeoff between accuracy and efficiency faced by the traditional Gradient Boosted Tree model via using gradient-based one-side sampling and exclusive feature bundling. In essence, this model uses a smaller data size to reduce the computation time without sacrificing the model accuracy. LightGBM is known for reducing forecasting biases and improving accuracy (Ke et al., 2017). There is limited literature on using xAI for cryptocurrency return prediction, but machine learning models have shown high performance in predicting Bitcoin prices in the short to medium term (Mudassir, 2020). SHAP values provide interpretable insights into individual uncertainty factors' impact on cryptocurrency returns. Gramegna & Guidici (2021) suggest that SHAP values are more suitable than LIME to be divided into clusters in the unsupervised settings.

3. Data and Methodology

The study utilizes a comprehensive dataset that includes both China and Japan EPU, and global uncertainty factors. The daily global uncertainty data and cryptocurrency data (BGCI) were obtained from Bloomberg, while the EPU data were sourced from the policy uncertainty database (Baker et al. 2016). China (CHE) and Japan (JEP) represent Asian countries⁹. China and Japan are selected because they represent the two largest macro-policy anchors in Asia and together account for the largest share of the regional digital economy. Uncertainty measures from other major Asian economies (Singapore, India and South Korea) were examined during the M–G causality test but

⁹ We collected datasets for other major Asian countries, including India, Singapore, and South Korea. However, neither the empirical nor theoretical evidence supports the notion that their policy related uncertainty has a significant causal relationship with cryptocurrency.

did not exhibit statistically significant nonlinear predictive content for cryptocurrency returns, a result that is consistent with Cheng and Yen (2020) and is therefore not reported in detail. BGCI is a market-capitalization-weighted benchmark designed to track the performance of the largest 11 cryptocurrencies traded in U.S. dollars ¹⁰. Bitcoin and Ethereum together account for approximately 70% of the index, broadly mirroring their combined share of total cryptocurrency market capitalization.

The global uncertainty data cover gold volatility (GOLD), oil volatility (OIL), the infectious disease index (IDV), CBOE volatility (VIX), geopolitical risk (GPRD), the world uncertainty index (WUI), and the U.S. EPU (USEPU). The EPU dataset was transformed from monthly to daily intervals to maintain data frequency consistent. This frequency alignment follows established method in the EPU–cryptocurrency studies (e.g., Cheng & Yen, 2020; Demir et al., 2018). To mitigate concerns that frequency alignment may introduce excessive short-term noise, the empirical specification uses rolling-window EPU correlations that can smooth high-frequency fluctuations and captures time-varying regional co-movement in policy uncertainty (Forbes & Rigobon, 2002; Baker, et al., 2016; Diebold & Yilmaz, 2012). Our dataset spans from September 1, 2017 to November 1, 2023. We collected a total of 15,113 daily observations. To ensure consistency in the database, we removed invalid data for IDV and retained 1,058 daily observations for each variable (T) across 10 variables (N). This results in a total of 10,580 daily observations ($T \times N$). Logarithmic returns for both cryptocurrency and the uncertainty indices were calculated based on price data.

¹⁰ Its eleven largest constituents were Bitcoin (35.00%), Ethereum (35.00%), Cardano (6.62%), Polygon (4.50%), Polkadot (3.7%), Litecoin (3.28%), Solana (3.15%), Uniswap (2.53%), Avalanche (2.38%), Chainlink (2.09%), and Cosmos (1.74%). The larger weight of each constituent asset in BGCI compares to S&P 500 or NASDAQ Composite reflects the dominant segment of the cryptocurrency market.

In our base analysis, we use rolling-window EPU correlations (Forbes & Rigobon, 2002) of China and Japan over horizons ranging from 30 to 240 days in order to account for the fact that policy uncertainty shocks may transmit into cryptocurrency markets with lagged and persistent effects rather than instantaneously. This rolling specification enables us to distinguish between short-lived uncertainty disturbances and longer systematic policy regimes, consistent with the uncertainty frameworks in Bloom (2009) and Baker et al. (2016).

Table 1 presents the summary statistics. All factors exhibit positive excess kurtosis, indicating leptokurtic distributions characterized by sharp peaks. Notably, VIX, OIL, and GOLD in Panel b, and BGCI in Panel d, show higher excess kurtosis. This suggests a greater likelihood of sudden spikes in returns due to political, economic, or health-related events, such as the significant impact of COVID-19. The negative skewness of BGCI and WUI suggests distributions with small and frequent gains but a small probability of large losses, highlighting a higher risk of downturns. The Jarque-Bera test results indicate significant deviations from normality, evidenced by the 1% level of significance, rejecting the null hypothesis of normally distributed returns.

Figure 1 graphically illustrates the overall return performance of uncertainty factors. The CHE and JEP indexes show relatively smaller fluctuations, whereas global uncertainty (VIX, OIL, WUI, IDV, GOLD, and GPRD) and U.S. EPU returns exhibit larger and more frequent spikes, particularly in OIL and VIX. During the COVID-19 pandemic, all plots show sharper spikes, suggesting that systematic risk significantly affected the markets.

Table 1: Descriptive Summary

	Mean	Maximum	Minimum	Std.	Skewness	Excess Kurtosis	Jarque-Bera
Panel a: China & Japan uncertainty							
CHE	0.0003	0.0630	-0.0481	0.0123	0.0294	2.8824	366.0608***

JEP	-0.0001	0.0555	-0.0491	0.0089	0.5912	3.5599	619.7303***
Panel b: Global uncertainty							
VIX	-0.0002	0.4802	-0.2662	0.0780	1.1026	4.5377	1121.0180***
OIL	-0.0001	0.8577	-0.6223	0.0752	2.0169	29.1626	38172.1700***
GOLD	-0.0003	0.3021	-0.2657	0.0520	0.8752	4.9432	1211.1090***
WUI	-0.0006	0.0583	-0.0844	0.0186	-0.5351	1.0585	99.7849***
GPRD	0.0021	2.3449	-1.6366	0.4483	0.0527	1.1469	58.4183***
IDV	0.0072	3.7296	-3.3397	0.8080	0.0116	1.7050	128.0522***
Panel c: U.S. uncertainty							
USEPU	-0.0020	2.2790	-1.9103	0.4940	0.1370	1.0026	47.5786***
Panel d: Cryptocurrency							
BGCI	0.0015	0.1983	-0.3361	0.0469	-0.9180	6.3435	1920.7030***

Notes: The data source for Economic Policy Uncertainty is the website *policyuncertainty.com*. The dataset consists of global uncertainty factors, U.S., China and Japan EPU's.¹¹ The number of observations for each variable is 1,058, summing up to 10,580 as a total. The monthly data was converted to daily data to maintain a consistent frequency across variables. Asterisks *** indicate two-tail statistical significance at 1% level.

3.1 M-G Model

As machine learning models alone are insufficient to identify directional predictive relationships among the variables, we use the bivariate M–G model (Mackey & Glass, 1977; Kyrtsov & Labys, 2006, 2007) to examine symmetric and asymmetric Granger predictive relationships between cryptocurrency returns and each uncertainty factor under a nonlinear specification. The model utilizes an algorithmic approach to estimate the parameters and examine whether past values of one variable contain significant predictive information for the current value of another variable under the nonlinear M–G specification. M-G model can avoid the regression distortions that occur from the combination of linear stochastic and chaotic series characteristics. To verify the asymmetric causality relationships, we test whether non-negative (non-positive) returns in BGCI (VIX, CHE, JEP, OIL, GOLD, GPRD, IDV, USEPU, WUI, SKE, INE and SEP) cause returns in VIX, CHE, JEP, OIL, GOLD, GPRD, IDV, USEPU, WUI, SKE, INE and SEP

¹¹ The descriptive summary for other major Asian countries is available upon request.

(BGCI), respectively. The M–G framework further allows us to examine whether positive or negative shocks contain significant predictive information for the other variable.

Equation (3.1) estimates the parameters that can best fit at the given time series and the causality effect of Y on X . On the other hand, Equation (3.2) tests the causality effect of X on Y .

$$X_t = \xi_{11} \frac{X_{t-\lambda_1}}{1 + X_{t-\lambda_1}^{\varphi_1}} - \alpha_{11} X_{t-1} + \xi_{12} \frac{Y_{t-\lambda_2}}{1 + Y_{t-\lambda_2}^{\varphi_2}} - \tau_{12} Y_{t-1} + \omega_t \quad \omega_t \sim N(0,1) \quad (3.1)$$

$$Y_t = \xi_{21} \frac{X_{t-\lambda_1}}{1 + X_{t-\lambda_1}^{\varphi_1}} - \alpha_{21} X_{t-1} + \xi_{22} \frac{Y_{t-\lambda_2}}{1 + Y_{t-\lambda_2}^{\varphi_2}} - \tau_{22} Y_{t-1} + \varrho_t \quad \varrho_t \sim N(0,1) \quad (3.2)$$

where λ_i are integer delays, while φ_i are constants that are established *a priori*. Both parameters, λ_i and φ_i are adjusted to generate dependence structures and dynamics between variables Y and X , so that we can distinguish M-G from the traditional linear Granger causality model. The M-G terms ξ_{ij} along with τ_{ij} are parameters to be estimated. The optimal delay λ_1 and λ_2 are determined by the Akaike information criterion (AIC)¹².

The nonlinear information and feedback flows between *BGCI* and each of *VIX*, *CHE*, *JEP*, *OIL*, *GOLD*, *GPRD*, *IDV*, *USEPU*, and *WUI* are confirmed by the rejection of the null hypotheses. To investigate whether these uncertainty factors have a predictive power on *BGCI*, (of the form, $X_{t-\lambda_1}/(1 + X_{t-\lambda_1}^{\varphi_1})$) on *BGCI* using Equation (3.1), we examine the null hypothesis, $\xi_{21} = 0$. Likewise, a predictive power of *BGCI* on each uncertainty can be observed by setting $\xi_{12} = 0$ with Equation (3.2). Furthermore, we assess their asymmetric causality relationships by testing non-negative (non-positive) returns in *BGCI* (*VIX*, *CHE*, *JEP*, *OIL*, *GOLD*, *GPRD*, *IDV*, *USEPU* and

¹² The parameters of optimal delay and exponent orders are reported in Appendix B.

WUI) cause returns in *VIX*, *CHE*, *JEP*, *OIL*, *GOLD*, *GPRD*, *IDV*, *USEPU*, and *WUI* (*BGCI*), respectively. Accordingly, statistical significance under the M–G framework is interpreted as evidence of predictive dependence identified under the nonlinear specification, rather than as evidence that the underlying relationship is exclusively nonlinear.

3.2. LightGBM

Ke et al. (2017) propose LightGBM based on the Gradient Boosted Decision Trees model framework (Friedman, 2001). Since then, it has gained popularity as a machine learning approach to enhance forecasting (Gong et al. 2022). LightGBM aims to minimize the expected value of a loss function $L(y, f(x))$ and employs regression trees to estimate the final model. The estimation process can be described as follows,

$$F_T(X) = \sum_{t=1}^M f_t(X) \quad (3.3)$$

where $F_M(X)$ is the final regression tree model and M is the number of regression trees. The optional branch weight of each branch node μ_i and the extreme value of an additive trained form, Γ_M , can be defined as follows,

$$\mu_i = \frac{\sum_{\gamma \in I_i} d_\gamma}{\sum_{\gamma \in I_i} \beta_\gamma + \delta} \quad (3.4)$$

$$\Gamma_M = -\frac{1}{2} \sum_{i=1}^I \frac{(\sum_{\gamma \in I_i} d_\gamma)^2}{\sum_{\gamma \in I_i} \beta_\gamma + \delta} \quad (3.5)$$

where d_γ and β_γ represent the first and second order gradient statistics of the loss function. I_i is the sample set of branches i and δ is the regularization weight. Γ_M is viewed as the scoring

function to measure the quality of the tree structure. Assume I_L and I_R are the left and right branches, separately, the objective function of LightGBM is defined below,

$$\Theta = \frac{1}{2} \left[\frac{(\sum_{\gamma \in I_L} d_{\gamma})^2}{\sum_{\gamma \in I_L} \beta_{\gamma} + \delta} + \frac{(\sum_{\gamma \in I_R} d_{\gamma})^2}{\sum_{\gamma \in I_R} \beta_{\gamma} + \delta} - \frac{(\sum_{\gamma \in I} d_{\gamma})^2}{\sum_{\gamma \in I} \beta_{\gamma} + \delta} \right] \quad (3.6)$$

4. Empirical analysis

We evaluate the predictive power of China and Japan uncertainty factors, controlling global and U.S. uncertainties for cryptocurrency returns in the framework of a combination of time-series econometric methods and machine learning algorithms. We first test the dynamic symmetric and asymmetric Granger causality between cryptocurrency returns and each uncertainty factor using the M–G approach. We then implement the LightGBM model integrated with SHAP values to assess predictive performance and variable importance. Robustness is evaluated using a range of alternative machine learning models. Within the predictive framework, we identify the most influential uncertainty factors affecting cryptocurrency returns and quantify the individual contribution of each variable, thereby measuring both the direction and magnitude of its impact on cryptocurrency returns.

4.1. Results from M-G Model

The study employs the M–G model to examine Granger predictive relationships between cryptocurrency returns and each uncertainty factor under a nonlinear specification. The empirical analysis using the M–G model is used to identify directional predictive dependence among the variables. The M–G framework examines symmetric and asymmetric pairwise predictive relationships in a bidirectional manner. The model examines three hypotheses: (1) Cryptocurrency

(uncertainty) returns do not nonlinearly cause changes in uncertainty (cryptocurrency) returns; (2) Positive cryptocurrency (uncertainty) returns do not nonlinearly cause changes in uncertainty (cryptocurrency) returns; and (3) Negative cryptocurrency (uncertainty) returns do not nonlinearly cause changes in uncertainty (cryptocurrency) returns.

Table 2, Panel A, presents the evidence of testing causality between cryptocurrency and each global uncertainty factor, while Panel B shows the results of its connection with China and Japan EPU (CHE and JEP). We first investigate global uncertainty factors with significant causality effects, which can be considered for the baseline model. The statistically significant M–G results provide evidence of asymmetric predictive dependence between several uncertainty factors and BGCI, including GOLD, OIL, IDV, VIX, GPRD, and USEPU. Specifically, the M–G results identify significant predictive dependence from positive changes in GOLD, OIL, and GPRD to BGCI, while negative changes in IDV, VIX, USEPU, and GOLD also contain significant predictive information for BGCI. These global uncertainty factors, with statistically significant causal relationships, are included as benchmark variables to account for their impact on BGCI. It is notable that, in the symmetric case, return information flows from BGCI to GOLD, OIL, VIX, and GPRD. In both positive and negative asymmetric cases, information flows from BGCI to GOLD, OIL, IDV, VIX, GPRD, and USEPU. These results suggest significant bidirectional predictive linkages between cryptocurrency returns and several global uncertainty factors, including GOLD, OIL, IDV, VIX, and GPRD, under the M–G framework. USEPU exhibits significant unidirectional predictive dependence with cryptocurrency returns under the M–G framework. This finding is consistent with (Gill-de-Albornoz et al. 2024; Ali et al. 2023). We find no evidence of significant predictive dependence from WUI to BGCI under the M–G framework in either the symmetric or asymmetric case.

We next examine whether China and Japan EPU's contain significant predictive information for cryptocurrency returns under the M–G framework, providing the basis for their consideration in the augmented model. Our objective is to investigate whether adding the augmented model to the baseline model can generate predictive benefits for cryptocurrency returns. This reverse linkage is economically meaningful, as it indicates that information generated in cryptocurrency markets feeds back into policy uncertainty dynamics in China and Japan, plausibly through the same channels that the literature documents as connecting crypto markets to broader information flows: investor sentiment (Anamika et al., 2023), media coverage of digital-asset volatility (Naifar et al., 2023), and regulatory discussions surrounding crypto adoption (Shanaev et al., 2020). To our knowledge, however, the impact of regional EPU co-movement on cryptocurrency markets has not been examined in the existing literature. This observation motivates our subsequent use of the rolling 30- to 240-day China–Japan EPU correlation, under the premise that the predictive signal of regional uncertainty lies in the co-movement of policy regimes rather than in individual country's EPU level alone.

Table 2: Results of Mackey-Glass Model

Null Hypothesis	(1)	(2)	(3)
<u>Panel A: Global Uncertainty Factor</u>			
GOLD does not cause BGCI	0.0023 (0.9617)	*3.4958 (0.0621)	***17.0388 (0.0000)
BGCI does not cause GOLD	**4.4101 (0.0360)	***7.1174 (0.0079)	0.7256 (0.3947)
OIL does not cause BGCI	0.0039 (0.9500)	*2.8465 (0.0922)	1.5264 (0.2172)
BGCI does not cause OIL	**6.5045 (0.0109)	**6.5186 (0.0110)	***7.9129 (0.0051)
IDV does not cause BGCI	0.0000 (0.9986)	1.2145 (0.2709)	**3.8696 (0.0497)
BGCI does not cause IDV	0.7950 (0.3728)	0.8461 (0.3581)	2.3004 (0.1300)
VIX does not cause BGCI	0.9091 (0.3406)	1.5712 (0.2106)	**4.4019 (0.0364)

BGCI does not cause VIX	*2.9463 (0.0864)	***9.2696 (0.0024)	***22.6140 (0.0000)
GPRD does not cause BGCI	0.0000 (0.9764)	*3.4204 (0.0650)	0.2490 (0.6180)
BGCI does not cause GPRD	*3.7609 (0.0527)	0.1623 (0.6872)	*3.3673 (0.0671)
WUI does not cause BGCI	0.9525 (0.0331)	0.1004 (0.7515)	2.2516 (0.1341)
BGCI does not cause WUI	4.5551 (0.3293)	1.3888 (0.2391)	5.4484 (0.0200)
USEPU does not cause BGCI	1.1700 (0.2797)	0.3268 (0.5678)	**4.0280 (0.0453)
BGCI does not cause USEPU	1.3671 (0.2426)	0.4670 (0.4947)	1.9990 (0.1581)
<u>Panel B: China & Japan EPU</u>			
CHE does not cause BGCI	*2.9166 (0.0880)	0.0037 (0.9513)	0.2162 (0.6421)
BGCI does not cause CHE	**4.6611 (0.0311)	**5.5189 (0.0192)	0.5526 (0.4576)
JEP does not cause BGCI	0.0613 (0.8044)	0.0000 (0.9816)	*3.6638 (0.0562)
BGCI does not cause JEP	***8.0023 (0.0048)	*3.4633 (0.0633)	0.0953 (0.7577)

Notes: This table presents the F-values for both symmetric and asymmetric connectedness. It examines predictive dependence between cryptocurrency returns and the individual uncertainty factors under the M–G framework. Panel A shows the effect of cryptocurrency returns on global uncertainty factors, while Panel B displays the causality effect of cryptocurrency returns on China and Japan EPU¹³. The causality results for other major Asian countries including Singapore, India and South Korea are insignificant and therefore excluded from the table. The p-values are provided in parentheses. Asterisks *, **, and *** indicate significance levels of 1%, 5%, and 10% in two-tailed tests, respectively. (1) represents the symmetric case, (2) refers to the positive asymmetric case, and (3) represents the negative asymmetric case.

The M–G results identify predictive dependence between cryptocurrency returns and the individual China and Japan EPU indices, motivating the inclusion of CHE and JEP in the subsequent machine learning analysis. We emphasize that the purpose of Table 2 is to examine predictive relationships between cryptocurrency returns and individual uncertainty factors. The correlation CHEJEP, by contrast, is a constructed measure of the time-varying co-movement between the two EPU series rather than an individual uncertainty index. Accordingly, its

¹³ The causality results for other major Asian countries are available upon request.

incremental predictive contribution is examined separately in the subsequent machine learning analysis.

4.2. Results from LightGBM with SHAP

The study utilizes xAI models, specifically LightGBM with SHAP, to investigate whether adding the augmented model to the baseline model increases the predictability of cryptocurrency returns. SHAP provides interpretability by revealing the individual contribution of uncertainty factors to cryptocurrency returns as produced by machine learning algorithms. In the xAI models, cryptocurrency returns serve as the dependent variable, while the uncertainty features serve as independent variables. Table 3 displays the model accuracy results using explainable LightGBM approaches. The baseline model consists of VIX, OIL, GOLD, USEPU, IDV, and GPRD, while the augmented model includes JPE and CHE.

Based on the M–G results reported in Table 2, CHE and JEP are first added to the base model to form the augmented model. We then incorporate the rolling correlation CHEJEP at 30-, 60-, 90-, 120-, 180-, and 240-day horizons to examine whether the time-varying co-movement between China and Japan EPU provides incremental predictive information beyond the baseline uncertainty factors and the individual CHE and JEP measures. We use 80% of dataset for the in-sample testing, while 20% of dataset is used for the out-of-sample testing. The model is trained with a maximum of 100 boosting rounds. Early stopping is applied when model performance fails to improve over 10 consecutive iterations. This helps mitigate overfitting by terminating training once additional boosting rounds no longer enhance validation accuracy. To prevent look-ahead bias, all tuning is conducted using information available prior to each forecast period. Appendix A reports a representative hyperparameter configuration for the augmented model.

We evaluate the predictive contribution of regional EPU information sequentially. First, we compare the augmented model containing CHE and JEP with the base model. We then add the CHE–JEP rolling correlation at alternative horizons to determine whether regional EPU co-movement provides additional predictive information beyond the individual CHE and JEP measures. The augmented model is classified based on the regional correlations between Japan and China EPU over short-, intermediate- and long-term periods including 30-day, 60-day, 90-day, 120-day, 180-day, and 240-day horizons. In the forecasting context, the baseline model combined with the intermediate-term 180-day correlation from the augmented model produces the best model performance, with the lowest RMSE (0.0409) and lowest MAE (0.0301). This suggests that the regional uncertainty between Japan and China over a 180-day horizon can significantly improve the predictability on cryptocurrency. This finding provides new insights to the interlinkages of cryptocurrency and China and Japan EPUs that has not been exposed in the current studies. On the other hand, the lower RMSE and MAE values indicate higher accuracy in the xAI models. Among the various combinations of the baseline model with and without the augmented model, the worst performance occurs when the baseline model is used without the augmented model, generating the highest RMSE (0.0585) and highest MAE (0.0372).

In addition to RMSE and MAE, we report Gini Norm to provide complementary information on the distribution of predictive contributions across uncertainty factors. While RMSE and MAE measure forecasting accuracy, Gini Norm captures the degree to which predictive contributions are concentrated or balanced across the uncertainty factors. It decreases from 0.3272 for the augmented model to 0.2457 and 0.1970 for the 180-day and 240-day correlation specifications, respectively, indicating more balanced predictive contributions across the uncertainty factors.

In general, we find that model performance improves by adding the augmented model into the base model, reducing both RMSE and MAE. Table 3 demonstrates the correlations of Japan and China in the augmented models across time intervals from 30-day to 240-day horizons. The results support the finding of incremental predictability when the regional intermediate-term correlation of uncertainty between Japan and China over a 180-day horizon is added to the global and U.S. uncertainty features.

Table 3: Results of Model Accuracy

Uncertainty Feature	(1)	(2)	(3)
Panel A. Base and individual EPU specifications			
Base model	0.0585 (0.0343)	0.0372 (0.0248)	0.3103 (0.7050)
Base model and CHE	0.0576 (0.0330)	0.0367 (0.0239)	0.3583 (0.7456)
Base model and JEP	0.0584 (0.0350)	0.0370 (0.0254)	0.3253 (0.7038)
Augmented model (Base model, CHE and JEP)	0.0582 (0.0332)	0.0367 (0.0240)	0.3272 (0.7522)
Panel B. CHE–JEP correlation specifications			
Augmented model with 30-day corr.	0.0515 (0.0340)	0.0346 (0.0241)	0.3530 (0.7542)
Augmented model with 60-day corr.	0.0459 (0.0386)	0.0327 (0.0268)	0.2958 (0.6967)
Augmented model with 90-day corr.	0.0418 (0.0341)	0.0309 (0.0233)	0.2658 (0.8121)
Augmented model with 120-day corr.	0.0466 (0.0372)	0.0348 (0.0254)	0.2950 (0.7384)
Augmented model with 180-day corr.	0.0409 (0.0364)	0.0301 (0.0254)	0.2457 (0.7360)
Augmented model with 240-day corr.	0.0455 (0.0402)	0.0303 (0.0288)	0.1970 (0.6597)

Notes: This table presents the statistical results of xAI model accuracy for out-of-sample testing using 20% of the dataset. The base model includes VIX, OIL, GOLD, USEPU, IDV, and GPRD. The augmented model additionally includes CHE and JEP. The augmented models with rolling correlations additionally include the CHEJEP correlation measured over the corresponding 30-, 60-, 90-, 120-, 180-, or 240-day horizon.

The in-sample testing results are shown in parentheses. (1) represents RMSE, (2) indicates MAE, and (3) refers to Gini Norm. A lower Root Mean Squared Error (RMSE) value represents better performance for the LightGBM model. Mean Absolute Error (MAE) measures the magnitude of prediction errors without considering direction. The lower value of MAE is better in predicting. The Gini Norm assesses how uncertainty factors impact cryptocurrency returns. Lower Gini Norm values suggest more balanced contributions across the uncertainty factors, while higher values indicate a disproportionately large influence on returns. Our primary evaluation criterion for model performance is the RMSE value, as it represents the magnitude of predictive accuracy. A representative hyperparameter configuration for the augmented model with 180-day corr. is presented in Appendix A.

Furthermore, the study investigates global interpretability using SHAP values for the model with the dataset of the baseline model and the regional EPU correlations of the augmented model. Figure 2 visually displays the individual feature contributions using SHAP for predicting cryptocurrency returns at a global interpretability level. The values of individual contributions explain the differences between the predicted values and the average predicted values for VIX, CHEJEP, OIL, GOLD, GPRD, IDV, and USEPU, respectively.

Among these features, CHEJEP ranks as the third most important, showing significant contributions with both positive and negative interpretability. It is not surprising that VIX and OIL have the highest impact on cryptocurrency returns, given the volatile nature of the cryptocurrency market. Both CHE and JEP are less important than CHEJEP. This finding highlights the incremental benefits of the correlation of Japan and China on predicting cryptocurrency returns. JEP has the least impact, with most SHAP values close to zero. The ranking of feature importance shows VIX at the top, followed by OIL, CHEJEP, GPRD, GOLD, CHE, IDV, USEPU and JEP. A dense cluster of SHAP values for CHEJEP is centered around zero but spreads in both positive and negative directions, indicating that the 180-day correlation of the augmented model can both positively and negatively impact cryptocurrency returns. The density of points for JEP is very close to zero, indicating its minimal impact among the group of uncertainty factors.

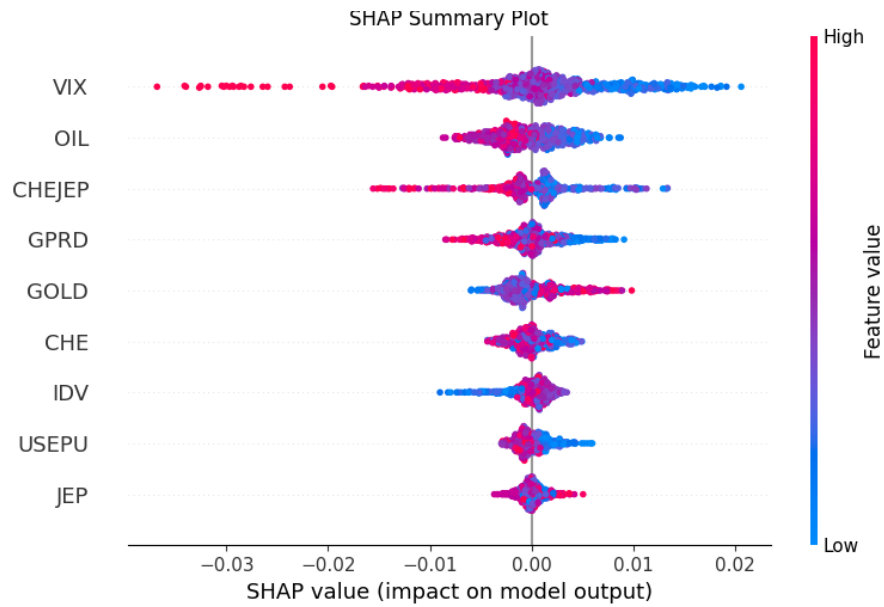


Figure 1: SHAP Value

Notes: The x-axis displays SHAP values, in which the positive SHAP suggests the feature pulling the cryptocurrency returns higher, while the negative SHAP indicates the feature dragging the cryptocurrency returns lower. The y-axis lists the feature importance in order, with the most important feature at the top of the plot and the least important feature at the bottom. Red color refers to the higher feature values while blue color indicates the lower feature values.

In Panel of Table 4, CHEJEP has the third highest aggregate absolute contribution (1.8342), following VIX (4.3978) and OIL (1.8709), and ahead of GPRD (1.4156), GOLD (1.4095), CHE (0.9171), IDV (0.8690), USEPU (0.7775), and JEP (0.5611). This suggests that the 180-day correlation of Japan and China EPU significantly impacts cryptocurrencies, contributing 13.07% of the total interpretability to the difference between predicted results and the average predicted value in the xAI algorithm. In Panel b, the feature relative impact of the 180-day correlation of Japan and China EPU (43.29%) is ranked as the third highest contributor to cryptocurrency returns, while IDV has the least impact (23.51%). However, a noteworthy finding is that the correlation between Japan and China, exert a greater impact on cryptocurrency returns compared

to other uncertainty factors, such as GPRD (41.25%), GOLD (38.66%), JEP (33.53%), USEPU (32.03%), CHE (26.56%), and IDV (30.48%). Panagiotidis et al. (2019) are unable to identify this meaningful correlation between Japan and China in predicting cryptocurrency returns, though they recognize the important role of Japan and China in their study. Our findings highlight the strong explanatory power of the 180-day correlation of Japan and China on cryptocurrency returns in comparison with various uncertainty factors in both regional and global levels.

Table 4: Results of Global Predictability from LightGBM

<u>Panel a: SHAP</u>			<u>Panel b: LightGBM</u>	
Feature	Aggregate Contribution	Weight	Feature	Relative Importance
VIX	4.3978	31.34%	VIX	100.00%
OIL	1.8709	13.19%	OIL	47.77%
CHEJEP	1.8342	13.07%	CHEJEP	43.29%
GPRD	1.4156	10.09%	GPRD	41.25%
GOLD	1.4095	10.04%	GOLD	38.66%
CHE	0.9171	6.54%	JEP	33.53%
IDV	0.8690	6.19%	USEPU	32.03%
USEPU	0.7775	5.54%	CHE	26.56%
JEP	0.5611	4.00%	IDV	23.51%
Total	14.0527	100%		

Notes: Table 4 reports feature importance for the base-model uncertainty factors, CHE, JEP, and the 180-day CHEJEP correlation. Panel A. shows the aggregate absolute SHAP values in explaining the outcome of LightGBM. Panel b reflects the total contribution of each feature to interpret cryptocurrency. The feature impact on cryptocurrency is ranked in Panel (b). The highest feature is ranked as 100%, suggesting the strongest impact on cryptocurrency. The importance is ranked in absention order. The higher values indicate a greater impact on the model's result. Incremental forecasting performance is evaluated separately through the model comparisons reported in Table 3.

Additionally, Figure 2 visually demonstrates the effect of each uncertainty factor on cryptocurrency returns. It graphically displays how the individual SHAP values change with varying uncertainty values for GOLD, GPRD, IDV, OIL, CHEJEP, USEPU, CHE, JEP and VIX. This helps identify both the linearity and nonlinearity of the explanatory power in describing how

individual uncertainty factors impact model predictions, thereby increasing the model's transparency and enhancing user trust. This downward linearity indicates a reduced effect on prediction outcomes, implying that higher uncertainty values lead to a diminished impact on cryptocurrency returns. There are downward trends in the plots for VIX, CHEJEP, OIL, CHE, and GPRD, suggesting that higher uncertainty values negatively impact their impact power on cryptocurrency returns. The spread of values in the CHEJEP plot suggests a consistent influence across different uncertainty levels.

In contrast, the plots for GOLD, IDV, JEP, and USEPU show nonlinear relationships between uncertainty and SHAP values. When uncertainty values are greater than around the zero-point, GOLD, JEP and USEPU exhibit positive trends, while IDV shows a downward trend. This suggests that increases in the feature values for GOLD, JEP and USEPU lead to a growing impact in explaining cryptocurrency returns, whereas increases in IDV reduce its predictive power. On the other hand, when uncertainty values are below the zero-point, GOLD, JEP and USEPU show downward trends, indicating that further increases in their values reduce their predictability. Conversely, the IDV plot shows an upward trend after the feature values surpass the zero point, suggesting a reverse relationship between predictability and feature value. These insights provide a deeper understanding of how each uncertainty factor contributes to the model's predictions and how changes in these uncertainties affect cryptocurrency returns.

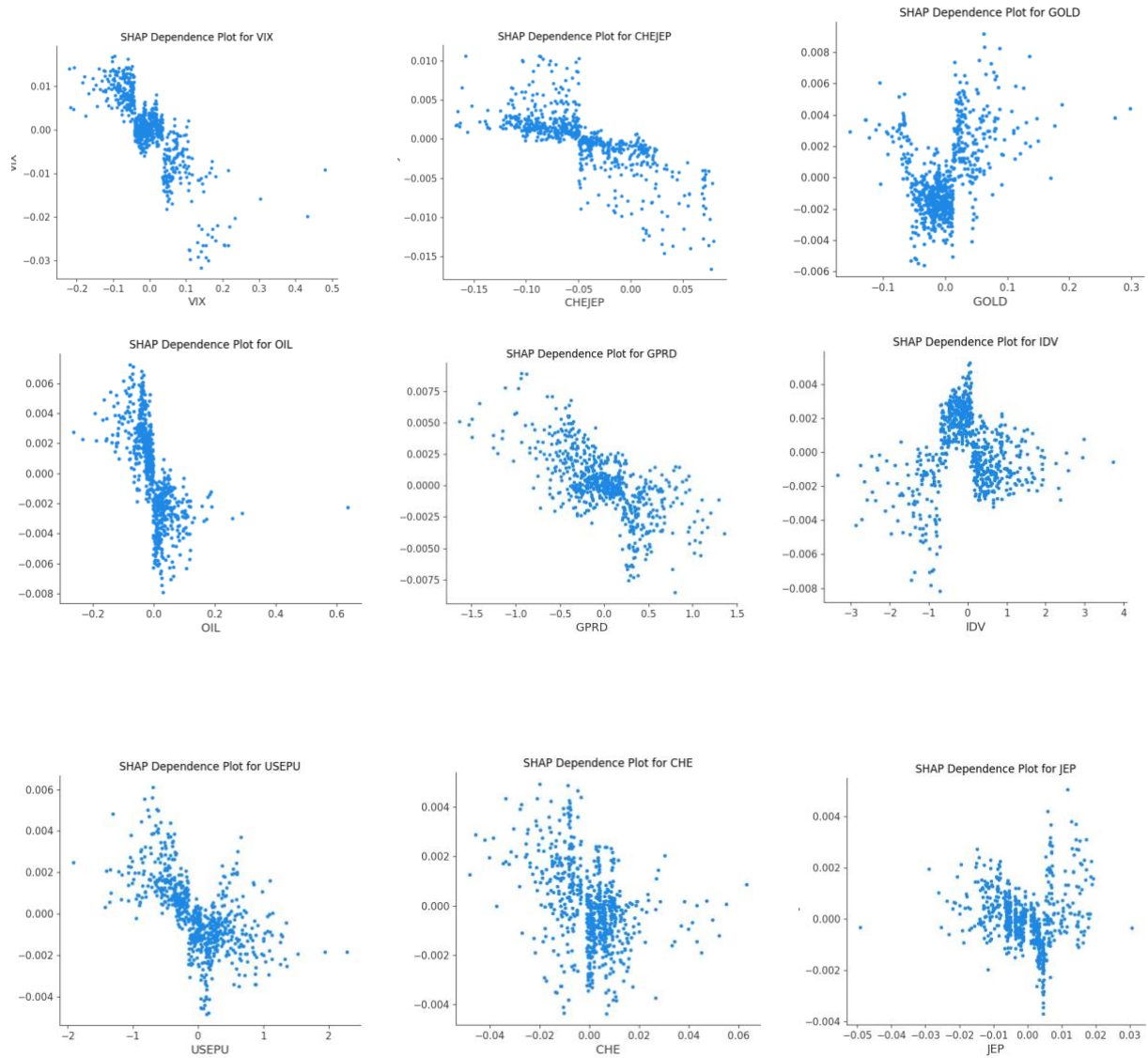


Figure 2: SHAP Dependent Plots

Notes: The x-axis indicates the actual values of the various uncertainty factors, showing the factor values vary across different time periods. The y-axis represents the SHAP values, indicating the magnitude of individual contribution of each uncertainty factor to the model outcome.

4.3. Robustness

Beyond the LightGBM algorithm, we aim to examine the robustness using alternative xAI models. Table 5 presents the robustness results, testing whether the incremental predictability of adding regional EPU correlations of the augmented model to baseline model on cryptocurrency returns exists across other machine learning models. The results are verified by both Random Forest (Panel a) and Ridge Regression (Panel b) models.

In alignment with the empirical questions using the LightGBM with SHAP approach, we assess model accuracy using the same dataset but with different methodologies. Panel a of Table 5 shows that the best-performing model using Random Forest is the one that includes the augmented model and regional uncertainty correlation between China and Japan over a 180-day horizon (RMSE: 0.0397; MAE: 0.0286), supporting the evidence from the LightGBM approach. The worst-performing model includes only the baseline model (RMSE: 0.0598; MAE: 0.0382), further confirming the findings from Table 3. The model predictability is improved by adding individual CHE and JEP into the baseline model (RMSE: 0.0581; MAE: 0.0368).

Similarly, the results in Panel b using Ridge Regression show robustness. The model that incorporates the regional intermediate-term correlation of uncertainty in China and Japan over a 180-day horizon (RMSE: 0.0400; MAE: 0.0289) has the lowest prediction errors for cryptocurrency returns. Conversely, model with only the baseline model produces the highest prediction errors (RMSE: 0.0576; MAE: 0.0359). Further, the model predictability is improved by adding individual CHE and JEP (RMSE: 0.0576; MAE: 0.0359). For both Random Forest and Ridge Regression, the 240-day correlation specification produces a lower Gini Norm than the 180-day specification, indicating more balanced contributions across the uncertainty factors. In contrast, the 180-day specification produces the lowest RMSE for both models, indicating better forecasting

accuracy. These results show that a more balanced distribution of predictive contributions does not necessarily correspond to higher forecasting accuracy.

It can be noticed that in-sample testing shown in parentheses generally exhibits less prediction errors, but inconsistent results compared to out-of-sample testing. This discrepancy may be due to overfitting, where the models capture noise or specific patterns in the training data that do not generalize well to unobserved data in the out-of-sample testing. Furthermore, in practice, investors use real-time information to make out-of-sample prediction. Therefore, we focus on comparing model accuracy based on the out-of-sample results.

Table 5: Robustness Results of Two Models Accuracy

Uncertainty Feature	Panel a: Random Forest			Panel b: Ridge Regression		
	(1)	(2)	(3)	(1)	(2)	(3)
Base model	0.0598 (0.0156)	0.0382 (0.0114)	0.2235 (0.9663)	0.0576 (0.0406)	0.0359 (0.0294)	0.3161 (0.3247)
Augmented model	0.0581 (0.0154)	0.0368 (0.0113)	0.2909 (0.9708)	0.0576 (0.0408)	0.0359 (0.0295)	0.3158 (0.3247)
Augmented model with 30-day corr.	0.0532 (0.0157)	0.0364 (0.0115)	0.2201 (0.9709)	0.0506 (0.0412)	0.0341 (0.0295)	0.3643 (0.3208)
Augmented model with 60-day corr.	0.0468 (0.0169)	0.3435 (0.0117)	0.2681 (0.9701)	0.0467 (0.0424)	0.0333 (0.0297)	0.2635 (0.3348)
Augmented model with 90-day corr.	0.0417 (0.0171)	0.0308 (0.0120)	0.2921 (0.9726)	0.0407 (0.0441)	0.0298 (0.0306)	0.4509 (0.3107)
Augmented model with 120-day corr.	0.0482 (0.0168)	0.0356 (0.0117)	0.2329 (0.9678)	0.0469 (0.0425)	0.0346 (0.0291)	0.2479 (0.3591)
Augmented model with 180-day corr.	0.0397 (0.0163)	0.0286 (0.0114)	0.3691 (0.9761)	0.0400 (0.0429)	0.0289 (0.0298)	0.3431 (0.3243)
Augmented model with 240-day corr.	0.0475 (0.0164)	0.0316 (0.0118)	0.2544 (0.9709)	0.0460 (0.0423)	0.0309 (0.0301)	0.1904 (0.3442)

Notes: This table presents the results of robustness. It compares model accuracy for Random Forest (Panel a) and Ridge Regression (Panel b) in out-of-sample testing using 20% of the dataset. The in-sample testing results are shown in parentheses. (1) represents RMSE, (2) indicates MAE, and (3) refers to Gini Norm. A lower RMSE value represents better performance of xAI model. MAE measures the magnitude of prediction errors without considering direction. The lower value of MAE is better in predicting. The Gini Norm assesses how uncertainty factors impact cryptocurrency returns. Lower Gini Norm values suggest

more balanced contributions across the uncertainty factors, while higher values indicate a disproportionately large influence on returns. Consistent with the interpretation in Table 3, The variation in Gini Norm across the specifications indicates differences in the concentration of predictive contributions among the uncertainty factors. Our primary evaluation criterion for model performance is the RMSE value, as it represents the magnitude of predictive accuracy.

5. Conclusion

Growing concerns about the impact of economic policy uncertainty on financial markets have intensified interest in understanding the role of uncertainty in cryptocurrency returns. This paper extends the existing literature by examining whether regional economic policy uncertainty interactions—specifically the correlation between China and Japan EPU—provide incremental predictive power beyond global uncertainty measures and U.S. EPU.

Our empirical results show that incorporating the regional EPU correlation between China and Japan significantly improves the out-of-sample predictability of cryptocurrency returns. The analysis framework is beyond the conventional correlation test (*Ozili, 2021*). In particular, we find that intermediate-term correlation over a 180-day horizon yields substantial forecasting gains relative to a baseline model. These findings highlight the importance of capturing cross-country uncertainty spillovers when modeling cryptocurrency markets.

Using explainable artificial intelligence techniques, we further demonstrate that the China–Japan EPU correlation contributes meaningfully to return forecasts, ranking third to VIX and OIL in predictive importance. This evidence underscores the role of regional uncertainty interactions in shaping cryptocurrency price dynamics. Overall, our study contributes to the growing literature on uncertainty and digital asset markets by identifying a novel regional spillover channel and offering practical implications for investors and policymakers seeking to assess risk in the evolving digital economy.

Several limitations of the present study suggest directions for future research. First, our empirical strategy relies on rolling-window correlations of EPU to mitigate concerns arising from the monthly-to-daily frequency conversion of the EPU series. Future work could explore alternative mixed-frequency techniques to extend this framework. Second, extending the analysis to more periods and to a broader cross-section of digital assets, including individual cryptocurrencies and stablecoins, would provide a useful extension for understanding cryptocurrency markets.

Appendix

A. LightGBM Hyperparameter Configuration

Hyperparameter	Description	Value
n_estimators	Number of boosting iterations (trees)	21
learning_rate	Step size shrinkage parameter	0.05
max_depth	Maximum depth of individual trees	-1
num_leaves	Maximum number of leaves per tree	31
min_data_in_leaf	Minimum observations per leaf node	20
subsample	Fraction of observations used for each tree	1
colsample_bytree	Fraction of features sampled per tree	0.9
objective	Loss function	RMSE

Notes: This table summarizes the full set of baseline hyperparameters for augmented model with 180-day correlation. The specification is relatively parsimonious. The early stopping results in effective number of 21 boosting iterations, indicating that the model does not need additional trees to improve forecasting performance. Model complexity is controlled through the leaf structure with 31 leaves and 20 observations per leaf. It indicates that model does not require an explicit depth restriction. This tuning strategy alleviates concerns regarding potential overfitting.

B: Prior Parameters for M-G Model

Factor	Symmetric				Asymmetric (P)				Asymmetric (N)			
	λ_1	λ_2	φ_1	φ_2	λ_1	λ_2	φ_1	φ_2	λ_1	λ_2	φ_1	φ_2
GOLD	4	6	10	2	2	9	2	3	1	9	1	4
OIL	1	6	9	1	1	10	3	1	2	2	10	2
IDV	2	10	2	1	5	9	4	1	6	10	2	1
VIX	1	6	2	1	3	10	1	1	10	2	4	1
GPRD	2	10	2	2	2	10	2	1	9	5	5	1
WUI	9	5	5	1	1	9	4	1	1	10	4	2
USEPU	2	10	2	1	5	10	1	1	8	9	7	2
CHE	1	9	2	1	8	10	1	10	1	10	8	3
JEP	1	1	6	10	1	9	7	3	9	10	2	1

Notes: λ_1 and λ_2 are the delay parameters, while φ_1 and φ_2 are the exponents in the M-G model. These parameters are selected *a priori* using the AIC to test for the symmetric, positive asymmetric and negative asymmetric specifications of the M-G model.

References

- Ahir, H., Bloom, N., & Furceri, D. (2022). The world uncertainty index (No. w29763). *National bureau of economic research*.
- Ali, M. H., Schinckus, C., Uddin, M. A., & Pahlevansharif, S. (2023). Asymmetric effects of economic policy uncertainty on Bitcoin's hedging power. *Studies in Economics and Finance* (Charlotte, N.C.), 40(2), 213–229.
- Almeida, J., & Gonçalves, T. C. (2024). Economic policy uncertainty and cryptocurrencies: A country-level wavelet analysis. *Journal of Alternative Investments*, 27(2), 90–113.
- Anamika, Chakraborty, M., & Subramaniam, S. (2023). Does sentiment impact cryptocurrency? *Journal of Behavioral Finance*, 24(2), 202–218.
- Aysan, A. F., Demir, E., Gozgor, G., & Lau, C. K. M. (2019). Effects of the geopolitical risks on Bitcoin returns and volatility. *Research in International Business and Finance*, 47, 511–518.
- Baker, S. R., Bloom, N., Davis, S.J., (2016). Measuring economic policy uncertainty. *The Quarterly Journal of Economics*, 131 (4), 1593–1636.
- Bariviera, A. F. (2017). The inefficiency of Bitcoin revisited: A dynamic approach. *Economics Letters*, 161, 1–4.
- Barmuta, K. A., Akhmetshin, E. M., Andryushchenko, I. Y., Tagibova, A. A., Meshkova, G. V., & Zekiy, A. O. (2020). Problems of Business Processes Transformation in the Context of Building Digital Economy. *Entrepreneurship and Sustainability Issues*, 8(1), 945.
- Bloom, N. (2009). The impact of uncertainty shocks. *Econometrica*, 77(3), 623–685.
- Bloom, N. (2014). Fluctuations in Uncertainty. *Journal of Economic Perspectives*, 28 (2), 153–176.
- Bloom, N., Floetotto, M., Jaimovich, N., Saporta-Eksten, I. & Terry, S. J. (2018). Really Uncertain Business Cycles. *Econometrica*, 86 (3), 1031–1065.

- Biljanovska, N., Grigoli, F., & Hengge, M. (2021). Fear thy neighbor: Spillovers from economic policy uncertainty. *Review of International Economics*, 29(2), 409–438.
- Boao Forum for Asia, & China Academy of Information and Communications Technology (CAICT). (2023). *Asia Digital Economy Report 2023*.
- Bouoiyour, J., Selmi, R., Tiwari, A. K., & Olayeni, O. R. (2016). What drives Bitcoin price. *Economics Bulletin*, 36(2), 843–850.
- Bouri, E., Gupta, R., Tiwari, A. K., & Roubaud, D. (2017). Does Bitcoin hedge global uncertainty? Evidence from wavelet-based quantile-in-quantile regressions. *Finance Research Letters*, 23, 87–95.
- Bussmann, N., Giudici, P., Marinelli, D., Papenbrock, J., (2021). Explainable Machine Learning in *Credit Risk Management*. *Computational Economics*, 57, 203–216.
- Calvo, G. A., & Vegh, C. A. (2010). Currency Substitution in Developing Countries: An Introduction. *Economic Analysis Review*, 7(1), 3–27.
- Cheah, E. T., & Fry, J. (2015). Speculative bubbles in Bitcoin markets? An empirical investigation into the fundamental value of Bitcoin. *Economics letters*, 130, 32–36.
- Cheng, H.P., & Yen, K.C. (2020). The relationship between the economic policy uncertainty and the cryptocurrency market. *Finance Research Letters*, 35.
- Chowdhury, E. K., & Abdullah, M. N. (2024). Gauging demand for cryptocurrency over the economic policy uncertainty and stock market volatility. *Computational Economics*, 64(1), 37–55.
- Chiew, E., & Choong, S. S. (2022). A solution for M5 Forecasting-Uncertainty: Hybrid gradient boosting and autoregressive recurrent neural network for quantile estimation. *International Journal of Forecasting*, 38(4), 1442–1447.
- Ciaian, P., Rajcaniova, M., & Kancs, D. A. (2016). The economics of BitCoin price formation. *Applied economics*, 48(19), 1799–1815.
- Civelli, A., & Jackson, L. E. (2025). Cryptocurrencies, stocks, and economic policy uncertainty: A FAVAR analysis. *The North American Journal of Economics and Finance*, 78, 102405.
- Corbet, S., Lucey, B., & Yarovaya, L. (2018). Datestamping the Bitcoin and Ethereum bubbles. *Finance Research Letters*, 26, 81–88.
- Demir, E., Gozgor, G., Lau, C. K. M., & Vigne, S. A. (2018). Does economic policy uncertainty predict the Bitcoin returns? An empirical investigation. *Finance Research Letters*, 26, 145–149.
- Diebold, F. X., & Yilmaz, K. (2012). Better to give than to receive: Predictive directional measurement of volatility spillovers. *International Journal of Forecasting*, 28(1), 57–66.
- Dyhrberg, A. H. (2016). Bitcoin, gold and the dollar—A GARCH volatility analysis. *Finance research letters*, 16, 85–92.
- Eberly, J. (1994). Adjustment of Consumers' Durables Stocks: Evidence from Automobile Purchases. *Journal of Political Economy*, 102 (3), 403–436.
- Eichengreen, B. (2019). From Commodity to Fiat and Now to Crypto: What Does History Tell Us? (2019). *NBER Working Paper Series*, 25426.
- Ertz, M., & Boily, É. (2019). The Rise of the Digital Economy: Thoughts on Blockchain Technology and Cryptocurrencies for the Collaborative Economy. *International Journal of Innovation Studies*, 3(4), 84–93.
- Fang, J., Gozgor, G., Lau, C. K. M., & Seetaram, N. (2022). Does policy uncertainty affect economic globalization? An empirical investigation. *Applied Economics*, 54 (22), 2510–2528.
- Forbes, K. J., & Rigobon, R. (2002). No contagion, only interdependence: measuring stock market comovements. *The Journal of Finance*, 57(5), 2223–2261.

- Friedman, M. (1956). *The quantity theory of money: a restatement*. 1–21.
- Gill-de-Albornoz, B., Lafuente, J. A., Monfort, M., & Ordoñez, J. (2024). Bitcoin attention and economic policy uncertainty. *Finance Research Letters*, 60, 1–6.
- Giudici, P., (2018). Fintech Risk Management: A Research Challenge for Artificial Intelligence in Finance. *Frontier in Artificial Intelligence*. 1, 1–6.
- Glaser, F., Zimmermann, K., Haferkorn, M., Weber, M. C., & Siering, M. (2014). Bitcoin - Asset or currency? Revealing users' hidden intentions. *ECIS 2014 Proceedings - 22nd European Conference on Information Systems*.
- Gong, X., Guan, K., & Chen, Q. (2022). The role of textual analysis in oil futures price forecasting based on machine learning approach. *Journal of Futures Markets*, 42(10), 1987-2017.
- Gramegna, A., & Giudici, P. (2021). SHAP and LIME: An Evaluation of Discriminative Power in Credit Risk. *Frontiers in Artificial Intelligence*, 4, 752558–752558.
- Haq, I. U., Ferreira, P., Quintino, D. D., Huynh, N., & Samantreeporn, S. (2023). Economic policy uncertainty, energy and sustainable cryptocurrencies: investigating dynamic connectedness during the COVID-19 pandemic. *Economies*, 11(3), 76.
- Haq, I. U., Maneengam, A., Chupradit, S., Suksatan, W., & Huo, C. (2021). Economic policy uncertainty and cryptocurrency market as a risk management avenue: A systematic review. *Risks*, 9(9), 163.
- Hassan, M. K., Hasan, M. B., & Rashid, M. M. (2021). Using precious metals to hedge cryptocurrency policy and price uncertainty. *Economics Letters*, 206, 109977.
- Hayes, A. S. (2017). Cryptocurrency value formation: An empirical study leading to a cost of production model for valuing bitcoin. *Telematics and informatics*, 34(7), 1308-1321.
- He, C., Li, Y., Wang, T., & Shah, S. A. (2024). Is cryptocurrency a hedging tool during economic policy uncertainty? An empirical investigation. *Humanities and Social Sciences Communications*, 11(1), 1-10.
- Hoque, M. E., Soo-Wah, L., Uddin, M. A., & Rahman, A. (2023). International trade policy uncertainty spillover on stock market: Evidence from fragile five economies. *The Journal of International Trade & Economic Development*, 32(1), 104-131.
- International Monetary Fund. (2024, October). *Regional Economic Outlook: Asia and Pacific—Resilient Growth but Higher Risks*. Washington, DC: IMF.
- Jabeur, S. B., Khalfaoui, R., & Arfi, W. B. (2021). The effect of green energy, global environmental indexes, and stock markets in predicting oil price crashes: Evidence from explainable machine learning. *Journal of Environmental Management*, 298, 113511.
- Jiang, Y., Nie, H., & Ruan, W. (2018). Time-varying long-term memory in Bitcoin market. *Finance Research Letters*, 25, 280-284.
- Kang, W., Ratti, R. A., & Vespignani, J. (2020). Impact of global uncertainty on the global economy and large developed and developing economies. *Applied Economics*, 52 (22), 2392–2407.
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T. Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems*, 2017(30), 3146–3154.
- Kristoufek, L. (2013). BitCoin meets Google Trends and Wikipedia: Quantifying the relationship between phenomena of the Internet era. *Scientific reports*, 3(1), 3415.
- Kyrtsou, C. and Labys, W. C., (2006) Evidence for chaotic dependence between U.S. Inflation and commodity prices. *Journal of Macroeconomics*, 28:256–266.

- Kyrtsov, C. and Labys, W.C., (2007) Detecting positive feedback in multivariate time series: the case of metal prices and U.S. inflation. *Physica A* 377(1):227-229.
- Li, R., Li, S., Yuan, D., Chen, H., & Xiang, S. (2023). Spillover effect of economic policy uncertainty on the stock market in the post-epidemic era. *The North American Journal of Economics and Finance*, 64, 101846.
- Liu, N., & Gao, F. (2022). The world uncertainty index and GDP growth rate. *Finance Research Letters*, 49, 103137.
- Mackey, M.C., & Glass, L., (1977). Oscillation and chaos in physiological control systems. *Science*, 197, 287-289.
- Malladi, R. K., & Dheeriyaa, P. L. (2021). Time series analysis of Cryptocurrency returns and volatilities. *Journal of Economics and Finance*, 45(1), 75–94.
- Mudassir, M., Bennbaia, S., Unal, D., & Hammoudeh, M. (2020). Time-series forecasting of Bitcoin prices using high-dimensional features: a machine learning approach. *Neural Computing & Applications*, 1–15.
- Nadarajah, S., & Chu, J. (2017). On the inefficiency of Bitcoin. *Economics Letters*, 150, 6-9.
- Ngene, G. & Wang, J. (2024). Arbitrage Opportunities and Feedback Trading in Regulated Bitcoin Futures Market: An Intraday Analysis. *International Review of Economics and Finance*, 89(B), 743-761
- Nguyen Quang, B., Le, T. H., & Nguyen Phuc, C. (2022). Influences of uncertainty on the returns and liquidity of cryptocurrencies: Evidence from a portfolio approach. *International Journal of Finance & Economics*, 27(2), 2497-2513.
- Osei, P. M., Djimatey, R., & Adam, A. M. (2021). Economic policy uncertainty linkages among asian countries: evidence from threshold cointegration approach. *Discrete Dynamics in Nature and Society*, 2021(1), 6656176.
- Ozili P. K. (2021) *International Review of Applied Economics*, 35(5), 714–728.
- Panagiotidis, T., Stengos, T., & Vravosinos, O. (2019). The effects of markets, uncertainty and search intensity on bitcoin returns. *International Review of Financial Analysis*, 63, 220–242.
- Pástor, L., & Veronesi, P. (2013). Political uncertainty and risk premia. *Journal of Financial Economics*, 110(3), 520–545.
- Qian, L., Jiang, Y., Long, H., & Song, R. (2020). The roles of Economic policy uncertainty and the COVID-19 pandemic in the correlation between cryptocurrency and stock markets. *The Singapore Economic Review*, 1-30.
- Rapach, D. E., Strauss, J. K., & Zhou, G. (2013). International stock return predictability: What is the role of the United States? *The Journal of Finance*, 68(4), 1633-1662.
- Sarjana, S., Najib, M. A. A., & Khayati, N. (2021). Bibliometric Analysis to Encourage the Development of Digital Economy Scientific Studies. Conference Towards ASEAN Chairmanship 2023, pp. 14-21. *Atlantis Press*.
- Schaal, E. (2017). Uncertainty and Unemployment. *Econometrica*, 85 (6), November, 1675-1721.
- Shaikh, I. (2020). Policy uncertainty and Bitcoin returns. *Borsa Istanbul Review*, 20(3), 257–268.
- Shanaev, S., Sharma, S., Ghimire, B., & Shuraeva, A. (2020). Taming the blockchain beast? Regulatory implications for the cryptocurrency market. *Research in International Business and Finance*, 51, 101080.
- Singh, S., Bansal, P., & Bhardwaj, N. (2022). Correlation between geopolitical risk, economic policy uncertainty, and Bitcoin using partial and multiple wavelet coherence in P5+ 1 nations. *Research in International Business and Finance*, 63, 101756.

- Sovbetov, Y. (2018). Factors influencing cryptocurrency prices: Evidence from bitcoin, ethereum, dash, litcoin, and monero. *Journal of Economics and Financial Analysis*, 2(2), 1-27.
- Su, C. W., Qin, M., Tao, R., Shao, X. F., Albu, L. L., & Umar, M. (2020). Can Bitcoin hedge the risks of geopolitical events? *Technological Forecasting and Social Change*, 159, 120182.
- Sun, X., Liu, M., & Sima, Z. (2020). A novel cryptocurrency price trend forecasting model based on LightGBM. *Finance Research Letters*, 32, 101084.
- Tiwari, A. K., Jana, R. K., Das, D., & Roubaud, D. (2018). Informational efficiency of Bitcoin—An extension. *Economics Letters*, 163, 106-109.
- Ul Haq, I. (2023). Economic policy uncertainty, energy and sustainable cryptocurrencies: investigating dynamic connectedness during the COVID-19 pandemic. *Economies*, 11(3), 1–23.
- Urquhart, A. (2016). The inefficiency of Bitcoin. *Economics Letters*, 148, 80-82.
- Valletta, R. & Bengali, L. (2013). What's Behind the Increase in Part-time Work? *San Francisco Federal Reserve Bank Economic Letter*, August 26.
- Wang, C., Shen, D., & Li, Y. (2022). Aggregate Investor Attention and Bitcoin Return: The Long Short-term Memory Networks Perspective. *Finance Research Letters*, 49, 103143.
- Wang, D. N., Li, L., & Zhao, D. (2022). Corporate finance risk prediction based on LightGBM. *Information Sciences*, 602, 259-268.
- Wang, G. J., Xie, C., Wen, D., & Zhao, L. (2019). When Bitcoin meets economic policy uncertainty (EPU): Measuring risk spillover effect from EPU to Bitcoin. *Finance Research Letters*, 31.
- Wang, J. (2025). The connectedness between cryptocurrency and segmented technology sectors in different regulatory regimes: causality and machine learning approaches. *Applied Economics*, 1–17.
- Wang, J., & Micale, J. A. (2024). What happens when the disruptive asset meets the conventional asset? An analysis of cryptocurrency and REIT. *Applied Economics*, 1–14.
- Wang, J., & Ngene, G. M. (2020). Does Bitcoin still own the dominant power? An intraday analysis. *International Review of Financial Analysis*, 71, 1.
- Wang, M., Verma, R., Wang, J., Ngene, G., & Sylla, C. (2025). Do Global Uncertainty Factors Matter More to Cryptocurrency? *Journal of Risk and Financial Management*, 18(11), 628.
- World Bank. (2023). Digital Progress and Trends Report.
- World Bank. (2024). *World Development Indicators — GDP (current US\$): China, Japan*. Washington, DC: World Bank.
- World Bank. (2025). Digital Progress and Trends Report.
- Yin, L., Nie, J., & Han, L. (2021). Understanding cryptocurrency volatility: The role of oil market shocks. *International Review of Economics & Finance*, 72, 233-253.
- Zhang, P., Kong, D., Xu, K., & Qi, J. (2024). Global economic policy uncertainty and the stability of cryptocurrency returns: The role of liquidity volatility. *Research in International Business and Finance*, 67, 102165.
- Zhou, Y., Liu, Z., & Wu, S. (2022). The global economic policy uncertainty spillover analysis: In the background of COVID-19 pandemic. *Research in International Business and Finance*, 61, 101666.